Teaching Philosophy

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Teaching approach: from principles to evidence

My goal in teaching electrical and computer engineering is to help students connect a mathematical concept to a computational or physical system, evaluate how well that system works, and explain the reasoning behind an engineering decision. In communication networks and signal processing, for example, understanding a formula is only a starting point. Students should also be able to identify the assumptions in a channel or traffic model, design a repeatable experiment, interpret performance differences, and recognize where a model no longer describes reality. I seek to develop that progression through clear explanations, structured practice, and meaningful feedback.

My undergraduate teaching experience includes Computer Programming, Communication Systems, and Wireless Communication at the University of Lahore, where I also advised student projects and contributed to curriculum-related and assessment activities. At the University of Engineering and Technology, Taxila, I supported communications coursework, laboratories, assignments, and grading. At COMSATS University Islamabad, I mentored students on engineering research and thesis activities. These experiences showed me that students learn technical ideas more confidently when objectives are explicit, examples build progressively, and assignments offer opportunities to diagnose and correct errors.

Making theory practical and testable

I favor a sequence of conceptual explanation, worked example, guided activity, independent exercise, and reflection. For a lesson on packet transport or network reliability, students could first predict the effects of congestion, delay, or loss, then compare their predictions with a simple Python-based simulation, and finally explain when the results would differ on a real network. In a wireless communication topic, students might explore path loss, fading, interference, or link adaptation through parameterized experiments and connect those results to system-level reliability. These are proposed instructional examples, not claims about activities previously used in a particular course.

I also see programming as an engineering tool rather than an isolated requirement. Students learn more effectively when code helps them test a concept: tracing a communication protocol, visualizing a noisy signal, comparing regression models, or debugging an experiment. My research developing Python-based wireless network simulations and my industry experience analyzing telecommunications data provide realistic but accessible problems for such exercises. Depending on the course, I would use Python or appropriate existing tools rather than make unnecessary software demands on students.

Rutgers ECE's Computer and Communication Networks (14:332:423), Machine Learning for Engineers (14:332:443), and related communications and computational offerings align with my teaching interests and technical background. I would concentrate on assignments matched to the course syllabus and my expertise, collaborating with the course coordinator on prerequisites, laboratory activities, and assessment. I do not regard the ability to explain a research result as a substitute for preparing an undergraduate course thoroughly.

Assessment, feedback, and student engagement

I begin with observable learning outcomes, such as deriving a key relationship, explaining a protocol tradeoff, producing a correct implementation, or interpreting a dataset. I would align lectures, examples, homework, quizzes, and examinations with these outcomes. Short formative questions and small programming checkpoints can identify misconceptions before larger assignments are graded. For project work, a rubric should distinguish technical reasoning, implementation, validation, documentation, and communication, not only whether the final demonstration runs.

Feedback should help students understand what to improve next. I value annotated solutions, structured review sessions, timely office hours, and opportunities to revisit challenging ideas. For team assignments, I would establish individual responsibilities and assess both the technical result and each member's contributions. When students arrive with different programming, mathematical, or laboratory preparation, clear prerequisite reviews and carefully staged exercises can provide access to the material without lowering expectations.

Mentoring and continuous improvement

In supervising student projects and research, I aim to match the scope of a task to a student's preparation, beginning with a reproducible simulation or a well-defined analysis before advancing toward open-ended investigation. Students should learn to record assumptions, compare baselines, examine failures, and communicate results responsibly. My own work on network outages, measurement uncertainty, and machine-learning distribution shifts reinforces the importance of asking whether evidence actually supports a conclusion.

I would assess my effectiveness through student work, discussions, course feedback, and collaboration with Rutgers ECE colleagues. I am particularly interested in an instructional environment where undergraduate fundamentals and graduate-level engineering applications inform each other. My objective is for students to leave a course with not only new technical methods, but also the confidence and judgment to use them critically in real engineering situations.